1 Properties of Vector Spaces

Last time we defined a vector space to be a set V together with operations of addition and scalar multiplication which satisfy the following axioms:

Axioms of Addition
(A1) (Commutativity) $u+v=v+u$ for all $u,v\in V$.
(A2) (Associativity) $(u+v)+w=u+(v+w)$ for all $u,v,w\in V$.
(A3) (Additive Identity) There exists an element $0\in V$ such that $0+v=v$ for every $v\in V$.
(A4) (Additive Inverses) To each $v\in V$ there exists an element $-v\in V$ such that $v+(-v)=0$.

Axioms of Scalar Multiplication
(S1) (Associativity) $(\alpha\beta)v=\alpha(\beta v)$ for all $\alpha,\beta\in F$, $v\in V$.
(S2) (Distributivity over scalar addition) $(\alpha+\beta)v=\alpha v+\beta v$ for all $\alpha,\beta\in F$, $v\in V$.
(S3) (Distributivity over vector addition) $\alpha(v+w)=\alpha v+\alpha w$ for all $\alpha\in F$, $v,w\in V$.
(S4) (Multiplicative identity) $1v=v$ for all $v\in V$.

Since the axioms for addition are exactly the same as those of F, the proofs of the following properties are exactly the same as before:

Proposition 1.1.

(a) The zero vector $0\in V$ is unique.

(b) For each $v\in V$, its additive inverse $-v\in V$ is unique.

(c) The cancellation law holds in V: $u+v=w+v$ implies $u=w$.

We now prove various other properties which follow from the axioms of scalar multiplication.

Proposition 1.2.

(a) $\alpha 0 = 0$ *for all* $\alpha \in F$.

(b) $0v = 0$ *for all* $v \in V$.

(c) $-v = (-1)v$ *for all* $v \in V$.

(d) If $\alpha v = 0$, *then either* $\alpha = 0$ *or* $v = 0$.

1. Prove Proposition 1.2. Justify each step.